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Transfer and Uncertainty for Soil Spectroscopy: A Parsimonious Approach Using Bayesian Mixed Models and Soil Surveys

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ABSTRACT

Two challenges that limit the adoption of soil spectroscopy are (1) model transfer to settings with limited data, and (2) uncertainty quantification, especially with correlated errors. Linear mixed models (LMMs) have sucessfully addressed these challenges in other fields, but not in soil spectroscopy. Compared to machine learning transfer algorithms, LMMs are more straightforward and interpretable. LMMs can also leverage auxiliary variables, in our case: farm, field, and soil depth variables; and external soil survey (SSURGO) estimates of pH and texture. But instead of using auxiliary variables to rigidly stratify a dataset into groups and fit separate models, LMMs partially pool the entire dataset to balance local and global modeling. In this study, we predict soil carbon from field-moist NIR spectra of 459 samples of three soil layers (0–15, 15–30, 30–60 cm) in 155 cores across 18 agricultural fields of 8 farms in Northeastern USA (data and code provided). Using cross validation, we find that our LMM succeeds in model transfer between farms, outperforming existing approaches such as memory-based learning and Cubist. Moreover, as a probability model, the LMM automatically provides excellent uncertainty quantification, including for aggregated predictions where existing methods fail due to correlated errors. We conclude that LMMs show promise as a parsimonious approach for model transfer and uncertainty quantification in soil spectroscopy.

1 | Introduction

It is well established that the infrared spectrum of a soil can be used to predict various soil properties, including carbon concentration (Safanelli et al. 2025). While spectroscopy promises to reduce the processing and analysis costs for soil carbon and other properties (Sanderman et al. 2025), challenges remain that limit the uptake of this technology, including: (1) transfer, that is the performance of spectroscopy prediction models in soils with limited training data, and (2) uncertainty, that is the quantification of uncertainty in model predictions.

1.1 | Model Transfer

One way to improve spectroscopy model performance in a target soil (e.g., an agricultural farm) is to collect additional training data in the target soil or similar soils. However, this adds

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Highlights

- Transfer and uncertainty quantification are two major challenges for soil spectroscopy modeling.
- Linear mixed models (LMMs) have successfully addressed these challenges in other fields but not spectroscopy.
- We find LMMs outperform existing spectroscopy models in both transfer and uncertainty quantification.
- LMMs promise to address these challenges and make soil spectroscopy more cost-effective and robust.

costs for sampling, spectroscopic scans, and lab measurements. Alternatively, the spectroscopy model itself can be enhanced to improve performance on the target soil without new training data, and therefore without added cost. The performance of a model in different settings is called *model transfer* (Rossel et al. 2024).

A standard (baseline) model for soil spectroscopy is partial least squares regression (PLSR). This model specifies a single linear equation to predict a soil property from a spectrum. However, the underlying relationship between the spectrum and the target soil property may vary across different soils. Thus the transfer of a global model like PLSR can be hindered by the presence of training data from soils that are substantially different from the target soil.

One approach to model transfer, then, is to select a subset of available data for training, excluding dissimilar soils. Similarity can be estimated from the spectra themselves (e.g., memory-based learning, MBL) or by subsetting the data according to mapped soil types (e.g., model stratification). When selecting a subset, there is often a practical trade-off between its size (more training data is better) and its relevance (more relevant data is scarce) (Moura-Bueno et al. 2020).

Another approach to model transfer is to allow the relationship between the spectrum and the predicted soil property to vary non-linearly based on auxiliary information or the spectrum itself. This includes tree-based models such as Cubist and quantile regression forest (QRF). Because such models use all available training data, they potentially avoid the trade-off between training data size and relevance encountered by models that subset the data.

1.2 | Uncertainty Quantification

Uncertainty quantification helps us understand whether our spectroscopy predictions are accurate and precise enough to support a given task, such as mapping or mean-area estimation, and subsequent decision-making. Despite its importance, uncertainty quantification is a relatively under-explored area of soil spectroscopy (Wadoux and Ramirez-Lopez 2025). For example, most software implementations of PLSR do not quantify uncertainty.

We identified three main existing approaches to uncertainty quantification in the soil spectroscopy literature. The first is a set of techniques specific to PLSR (Zhang and Garcia-Munoz 2009). The second is quantile regression forest (QRF), which leverages the ensemble nature of random forest to quantify uncertainty (Meinshausen and Ridgeway 2006; Wadoux and Ramirez-Lopez 2025). The third is conformal prediction (Angelopoulos Anastasios and Bates 2023; Sanderman et al. 2025; Safanelli et al. 2025), a general framework for adding uncertainty quantification to any prediction model by estimating the "nonconformity" of new data with respect to the trained model.

Beyond making predictions for individual observations (soil samples), an important task is making aggregated predictions, for example, estimating mean soil carbon concentration across a field, farm, or other geography. This is relevant for national inventories as well as market and non-market carbon projects (Potash, Bradford, et al. 2025). While the "point prediction" of the mean is simply the mean of the point predictions of each individual observation, quantifying the uncertainty of this prediction is complicated because of correlated errors (Wadoux and Heuvelink 2025).

1.3 | This Study

In this study, we apply linear mixed models (LMMs) to soil spectroscopy. LMMs, also known as multilevel regression models, are widely applied in ecology (Bolker et al. 2009) and agronomy (Gbur et al. 2020) and are also used in soil science (Slaets et al. 2021; Makungwe et al. 2021). However, our review of the literature found just two studies using LMMs for soil spectroscopy modeling and neither of these investigated model transfer or uncertainty quantification (Wang et al. 2021; Martín-Sanz et al. 2025).

An LMM is a regression model in which some of the coefficients (intercept and slopes) vary across groups of observations according to a probability distribution. Moreover, LMMs can model this variation in coefficients by leveraging readily available auxiliary information. For example, in this study, we propose a specific LMM in which spectroscopy coefficients vary across agricultural farms, fields, and soil depth layers and are informed by field-level soil survey estimates of soil pH and texture.

LMMs have several important features that emerge from their definition. These features support their success across a wide range of applications and make them good candidates for both the model transfer and uncertainty quantification challenges of soil spectroscopy. In particular, the so-called "partial pooling" of observations across groups is a potential solution to the above trade-off between relevance and size of training datasets encountered by subsetting approaches to model transfer (Section 1.2). Additionally, LMMs use auxiliary information to make localized predictions in unobserved groups, for example, farms with no training measurements. Finally, LMMs can address uncertainty quantification because, as probability models, they automatically quantify the uncertainty of their predictions including both structural (parameter) uncertainty and residual uncertainty.

The purpose of this study is to investigate the potential of LMMs for soil spectroscopy. Specifically, we wish to assess LMM performance in comparison to existing approaches for the simultaneous challenges of (1) model transfer and (2) uncertainty quantification. We examine soil carbon concentration in a cluster of agricultural farms in Northeastern USA. This study is especially relevant to agricultural carbon inventories and markets, where spectroscopy has the potential to reduce the

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costs of measurement to make carbon projects more economical (Potash, Bradford, et al. 2025).

2 | Materials and Methods

2.1 | Data

The study was conducted in the Hudson Valley region of Northeastern USA (42 °N, 73 °W), at 18 fields belonging to 8 farms and located within 30 km of each other (Figure 1). The temperate climate has a mean annual temperature of 9°C and precipitation of 100 cm. The crops were a mix of corn-soy and hay and varied in size (3–12 ha) and elevation (60–280 m above sea level) (c.f. Polussa et al. 2026). For each field, we estimated soil pH and texture to 60 cm depth using area-weighted averages from the USDA NRCS Soil Survey Geographic Database (SSURGO); pH ranged from 5 to 6.5, and texture was predominantly silt loam (Figure 2).

We used spatial coverage sampling (Walvoort et al. 2010) to select 4 to 15 sampling locations per field (1 to 1.7 samples ha^{-1}). In winter 2023-2024 an intact soil core (4 cm diameter) was collected at each sampling location to a target depth of 60 cm using a truck-mounted hydraulic drill corer, or gas-powered handheld slide hammer where vehicular access was constrained. Cores were collected in plastic liners that were capped and refrigerated.

In the lab, field moist cores were extracted into a trough (split 10 cm diameter PVC pipe) and separated into depth layers of 0–15, 15–30, and 30–60 cm. Each layer was manually broken up and large rocks and roots removed. Spectroscopy scans were collected for each sample (depth layer) using a Neospectra

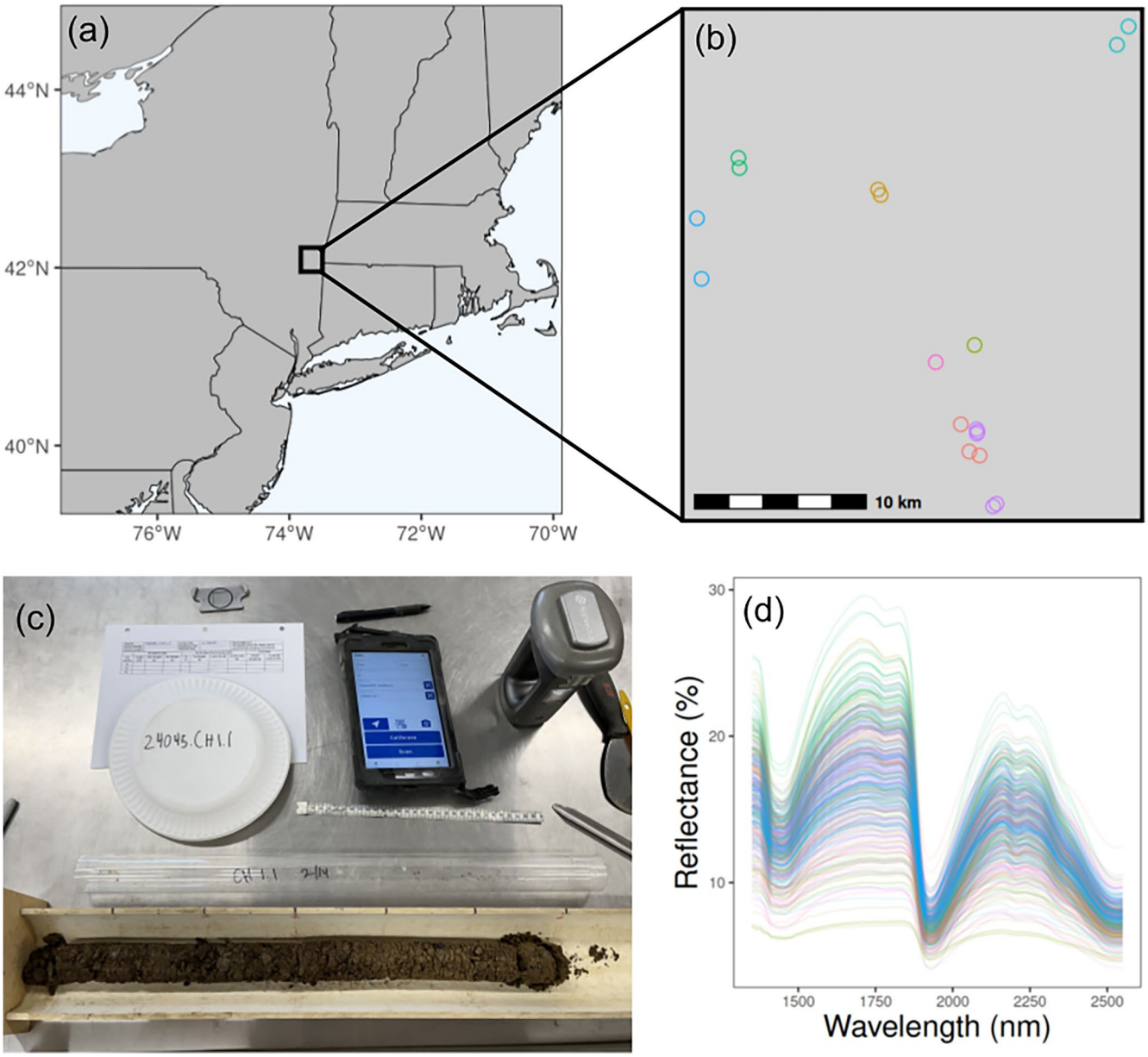


FIGURE 1 | Study design. (a) Study region of Northeastern USA. (b) Locations of 18 fields belonging to 8 farms (color) sampled for the study. (c) Handheld spectrometer scanning moist soil samples. (d) NIR reflectances for all 459 samples color-coded by farm. NIR, near infrared.

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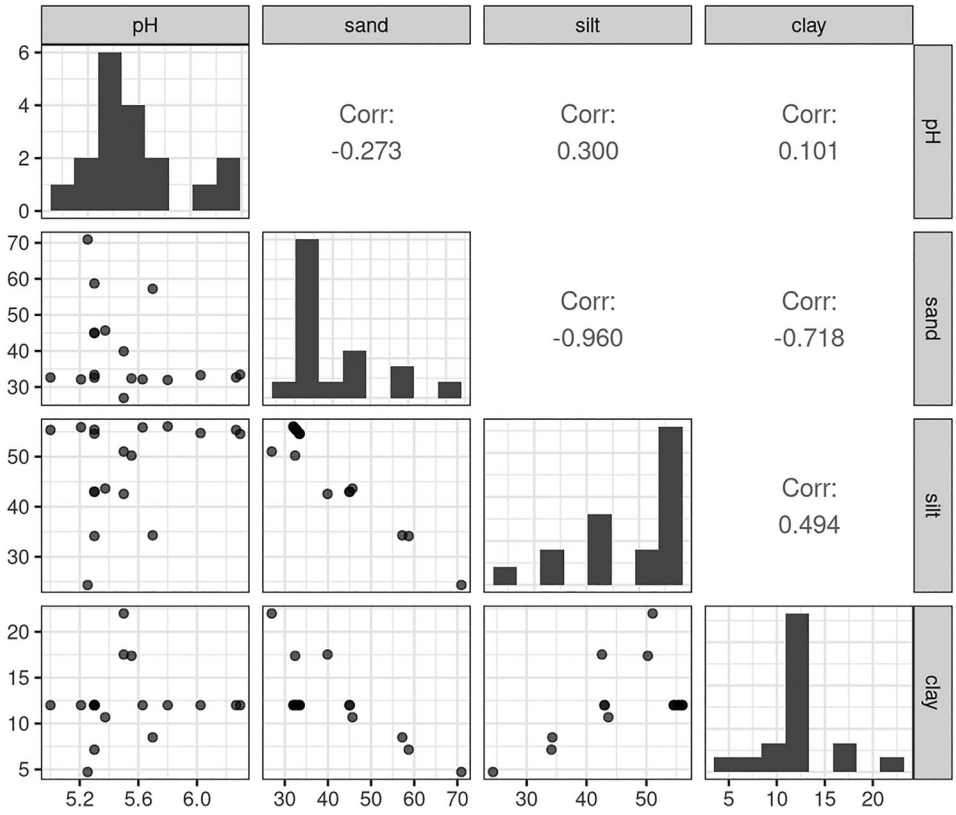


FIGURE 2 | Average 0-60 cm soil pH and texture properties estimated from the National Soil Survey (SSURGO) of the 18 study fields.

Scanner Handheld NIR Analyzer (Si-Ware, Egypt; now BUCHI Proxiscout) to measure NIR reflectances at 257 wavelengths between 1350 and 2550 nm. Six replicate scans were performed on each sample and averaged. Standard normal variate (SNV) transformation was applied to the averaged spectra. For the LMM, spectra were downsampled prior to SNV transformation, see Section 2.2.2.

The same samples were then air-dried and passed through a 2 mm sieve. Carbon concentration (%) was measured by dry combustion with an elemental analyzer (Vario MAX cube, Elementar Analysensysteme GmbH, Germany). According to their classification, these soils have negligible inorganic carbon, so total carbon concentrations are likely to reflect organic carbon concentrations. Carbon concentrations were modeled on the log1p scale (Safanelli et al. 2025).

2.2 | Models

An important feature of spectroscopy modeling is the relatively large number of predictors. Traditional regression typically does not work well when the number of predictors (here $n = 257$) is large relative to the number of observations (here $n \leq 459$). As a result, dimensionality reduction and regularization are common in spectroscopy modeling.

In the language of LMMs, our data have a multilevel structure of farms ($n = 8$), fields ($n = 18$), soil layers ($n = 3$), and field-layers ($n = 54$). We refer to any particular farm, field, soil layer, or field-layer as a group. A group-level predictor is a predictor that is constant within each group. In particular, the texture and pH variables described above are field-level predictors. We refer to a model that is constant between groups as a global model. When a model varies between groups we refer to the model in a particular group as a local model.

2.2.1 | Existing Models

Partial least squares regression (PLSR) is a classic model for spectroscopy (Geladi and Kowalski 1986) that combines regression with dimensionality reduction, accounting for the covariance between predictors and outcomes (Barra et al. 2021). We adopted the uncertainty quantification method of Faber and Kowalski (1996), which performed best in the evaluation of Zhang and Garcia-Munoz (2009). PLSR provides a global linear regression, though this may be transferred to a given prediction set (e.g., an agricultural farm). One approach to transfer is to stratify the training data based on auxiliary information. In this study, we stratified on soil texture (loam, sandy loam, or silt loam). However, stratification rigidly partitions the training data, creating a trade-off between training set relevance and training set size (Section 1.1).

Another approach to PLSR model transfer is memory-based learning (MBL). MBL searches for a subset of samples (neighbors) from the training set before fitting a prediction model (weighted average PLSR) for each sample to be predicted (Ramirez-Lopez et al. 2013). We implemented MBL using the `resemble` R package. The number of neighbors and number of PLSR components were selected by leave-nearest-neighbor-out cross-validation (`validation_type = "NNv"`). The dissimilarity matrix was

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included as a predictor in the model (`diss_usage = "predictors"`). Spiking was used to ensure that local observations, when available, were included as neighbors.

An alternative to dimensionality reduction is regularization, a feature of ensemble methods. Cubist is a tree ensemble with linear regressions at its leaf nodes. Random forest is another tree ensemble where a leaf node prediction is simply the average of the training observations in that node. Both Cubist and random forest inherently support model transfer because they are non-linear models with predictions that are localized to different regions of the spectroscopy input space. Cubist is a standard model in soil spectroscopy following Minasny et al. (2013). Random forest has been advanced for soil spectroscopy because of quantile random forest (QRF), a variant that quantifies uncertainty (Wadoux and Ramirez-Lopez 2025). Sanderman et al. (2025) quantified uncertainty for Cubist using conformal prediction.

2.2.2 | Linear Mixed Model (LMM)

We refer the reader to textbooks such as Gelman and Hill (2007) for a general introduction to LMMs and to Section 1.3 for their pertinence to model transfer and uncertainty quantification. In this study, we develop a specific LMM suitable for modeling our spectroscopy data as follows. Our first-level model is

$$\log(1+y_i) = X_i \beta_{\text{farm}[i],\text{field}[i],\text{layer}[i]} + \epsilon_i \tag{1}$$

where soil samples are indexed by i; y_i is the scalar soil carbon percentage from 0 to 100; X_i is the $K+1$ dimensional row vector spectral input for observation i (including 1 for an intercept); ϵ_i is a scalar residual; and $\beta_{\text{farm}[i],\text{field}[i],\text{layer}[i]}$ is a $K+1$ dimensional vector of regression coefficients that varies by the farm, field, and soil layer to which the observation belongs.

Specifically, we have the second-level regression

$$\beta_{\text{farm}[i],\text{field}[i],\text{layer}[i]} = \alpha + Z_{\text{field}[i]}\gamma + \nu_{\text{farm}[i],\text{field}[i],\text{layer}[i]} \tag{2}$$

where the data $Z_{\text{field}[i]}$ is an $L=4$ dimensional row vector of field-level predictors, that is the soil survey properties (pH, sand, silt, and clay) which were selected because they summarize soil chemical and physical properties and are known to be strongly predicted by the NIR spectrum (Safanelli et al. 2025). Then the fixed effects are α, a $K+1$ dimensional vector of second-level intercepts, and γ, a $(K+1)\times L$ matrix of second-level slopes which specify how the field-level predictors affect the first-level coefficients. Finally, the random effects $\nu_{\text{farm}[i],\text{field}[i],\text{layer}[i]}$ are a $K+1$ vector which is the residual in the second-level regression and are decomposed into field, layer, and field-layer components.

$$\nu_{\text{farm}[i],\text{field}[i],\text{layer}[i]} = \nu_{\text{farm}[i]} + \nu_{\text{field}[i]} + \nu_{\text{layer}[i]} + \nu_{\text{field}[i],\text{layer}[i]} \tag{3}$$

This multilevel structure facilitates sharing of information across fields of a given farm, layers of a given field, and fields of a given layer. Distributional assumptions for the residual ϵ and random effects ν are given in the supplement. Note that the statistical model (Equations 1–3) can be written in R formula syntax as:

$$\text{log1p(y)} \sim 1 + \text{Z} + \text{X} + \text{Z:X} + (1 + \text{X} \mid \text{farm} + \text{field} + \text{layer} + \text{field:layer})$$

where Y is carbon concentration, Z are the $L=4$ field-level predictors and X are the $K=64$ NIR predictors.

Traditionally LMM parameters have been estimated using restricted maximum likelihood (REML). However, Bayesian LMMs offer several advantages, and advances in computation have made them more practical (Gelman et al. 2013). To handle the large number of predictors in spectroscopy, a Bayesian LMM can impose regularization via shrinkage priors (ibid, section 14.6). For example, for the main fixed effects (spectroscopy slopes) we have a hierarchical prior:

$$\alpha_k \sim Normal\left(0, {\sigma_\alpha}^2\right) \quad k = 2, \ \ldots, K$$

where σ_α controls the degree of regularization and is given the weakly informative standard half-normal prior.

$$\sigma_\alpha \sim Normal_+(0,1)$$

Note that all priors are defined after standardizing both predictors X and outcomes y. In general our priors are chosen to be weakly informative, to aid computation and identifiability. For example, standard normal priors on the slopes of the field-level predictors ($\gamma_{1,l}$ for $l=1, ..., L$) allows us to include all three texture variables in the model despite the fact that these variables are collinear, always summing to 100%. See the supplement for the full prior specification and an assessment of prior informativeness (Figures S1–2).

We fit the Bayesian LMM using Markov Chain Monte Carlo (MCMC) which captures uncertainty more fully than REML. Let θ denote the vector of model parameters. Then MCMC generates a matrix of draws $\left\{\theta^{(s)}\right\}$ for $s=1, ..., S$ where $S=400$ is the number of posterior draws. We used the probabilistic programming language Stan (Carpenter et al. 2017) via the brms R package (Bürkner 2017). Our convergence diagnostics are described in the supplement. For computational reasons, we used cubic splines to interpolate the original 257 wavelengths (followed by SNV) into $K=64$. Data and R source code are provided.

Soil carbon predictions for a set of new soil samples indexed by $i'=1, ..., I'$ take the form of S draws $\left\{\hat{y}^{(s)}\right\}$ from the joint posterior predictive distribution where each draw $\hat{y}^{(s)}$ is a vector of predictions of length I'. To predict aggregated quantities, such as mean soil carbon concentration across the set of samples in a field-layer, we aggregate each draw $\hat{y}^{(s)}$ resulting in a set of posterior predictive draws for the aggregated quantities. Finally, point predictions and uncertainty intervals are generated from posterior predictive draws using the median and quantile functions, respectively.

2.3 | Cross Validation Scenarios

We used cross validation (CV) to evaluate transfer and uncertainty quantification under two scenarios: when no local observations (i.e., soil carbon measurements) are available and when some local observations are available. These scenarios emulate model transfer to a target farm with substantial amounts of

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training data from similar, nearby farms but little or no training data from the target farm.

Under "without local observations" CV, training included all samples outside the target farm (7 farms, 13–17 fields, 339 to 429 samples), and testing included all samples in the target farm (1–5 fields, 30 to 120 samples). Under "with local observations" CV, training additionally included 3 random cores (3 depths each, 9 samples total) in each field of the target farm (18 fields; 384 to 438 samples), and testing consisted of the remaining samples in the target farm (21 to 75 samples). "Without local observations" is akin to a "global" model and "with local observations" is akin to a "spiked" model (Ng et al. 2022). In both scenarios, auxiliary data (pH and texture) were used by the LMM for all fields.

For each CV fold, we fit the models (Section 2.2) on the training set to predict carbon concentration. We made point predictions for each sample in the test set. For models with uncertainty quantification (all except MBL), we generated 95% intervals for each of the test samples. Finally, we generated 95% intervals for the predicted mean carbon concentration across the finite set of samples in each field-layer in each test set. Note that the LMM jointly models uncertainty across samples. Other models produced intervals using statistical formulas that assume that prediction errors are uncorrelated (supplement).

For each of the eight folds of each CV scenario and each model we evaluated point prediction performance using the following metrics:

- Root mean-squared error (RMSE): $\sqrt{\mathrm{E}\left[(\hat{y}-y)^2\right]}$
- Ratio of performance to deviation (RPD): $\mathrm{sd}(y)/\mathrm{RMSE}$
- Absolute bias: $\left|\mathrm{E}[\hat{y}-y]\right|$
- Coefficient of determination (R^2): $\mathrm{cor}^2(y,\hat{y})$

Prediction intervals were evaluated by their empirical coverage rate (proportion that contained the measured value, ideally 95%). Because of the small size of the test sets, these were pooled into a single coverage rate under each scenario. Finally, we evaluated interval width as well as coverage using Winkler scores (Winkler 1972; supplement).

3 | Results

3.1 | Soil Carbon Distributions

Among the 459 soil samples, mean carbon was 1.2% with a standard deviation of 0.8%. We used a random-effects model to decompose the carbon distribution into six sources of variation (Figure 3c): layer, farm, field, core, field × layer, and sample (i.e., residual). The largest source of carbon variability was the soil layer (Figure 3a).

3.2 | Model Transfer

Without local observations, PLSR performance had mean (across eight folds, one for each farm) RMSE 0.49%, RPD 1.7, absolute bias 0.19%, and R^2 0.69. With local observations, these metrics improved to 0.40%, 2.0, 0.11%, and 0.74, respectively. We note that variability in lab measurements distorts estimates of prediction performance (Potash and Steinschneider 2022); however, dry combustion variability (estimated using replicate analyses to be 0.04%) is small relative to prediction error.

When comparing models, we observed the following patterns across metrics and CV scenarios (Figure 4). First, the stratified (PLSR) model did not consistently outperform the PLSR baseline, and QRF substantially underperformed it. Second, both MBL and Cubist tended to improve on PLSR, though the benefit was greater without local observations. Finally, the LMM performed best.

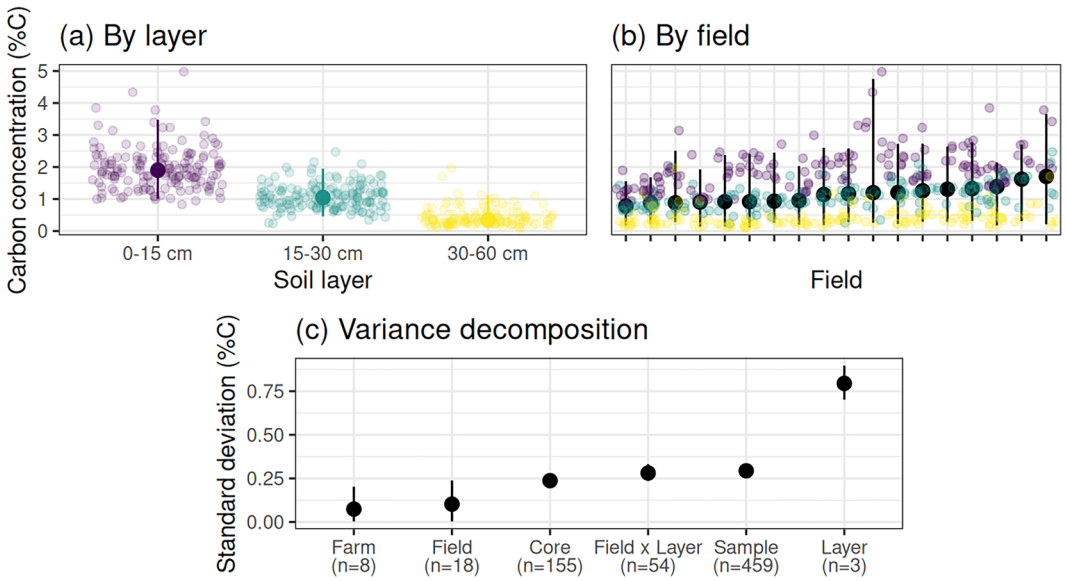


FIGURE 3 | Soil carbon statistics. (a) Distributions of raw measurements by soil layer. (b) Distributions of raw measurements by farm ($n = 8$), field ($n = 18$), sorted by median (dot) with 95% interval (line). (c) Variance decomposition from a random effects model.

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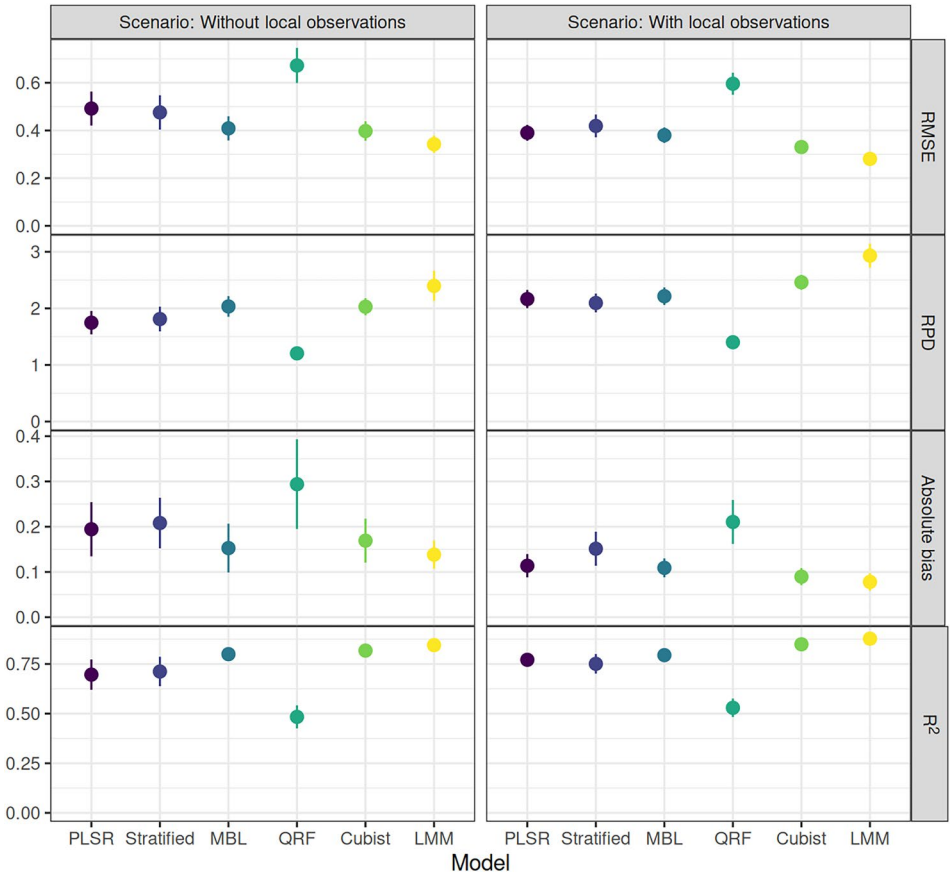


FIGURE 4 | Model transfer performance. Each model is evaluated under two transfer scenarios (Section 2.3): Without local observations and with local observations. Each model's performance in each scenario consists of 8 cross validation folds which are visualized by a point (mean) and bar (standard error). Abbreviations: Root mean squared error (RMSE), ratio of performance to deviation (RPD), coefficient of determination (R^2).

3.3 | Uncertainty Quantification

We generated 95% prediction intervals for each model under both transfer scenarios (without local observations, with local observations) at both the sample ($n = 459$) and field-layer ($n = 54$) levels. All models except MBL had individual sample-level uncertainty quantification and generated 95% intervals for field-layer means, whether directly (LMM) or through a standard formula (other models).

We first evaluated these intervals by comparing their empirical coverage to the nominal 95% (Figure 5). At the sample level, all models had close to nominal coverage. At the field-layer level and under both transfer scenarios, all models except LMM significantly underestimated their prediction uncertainty, with coverage rates between 60% and 80%. We note that the extent of this undercoverage is a function of model bias that is not accounted for in standard errors (supplement). Meanwhile, the LMM's posterior predictive distribution correctly produced intervals with approximately 95% coverage. The Winkler score evaluates both interval coverage and width and also shows that the LMM performs best, followed by Cubist (Figure S3).

3.4 | Linear Mixed Model Interpretation

The coefficients of the NIR reflectances (Figure 6a) align with prior studies of soil carbon spectroscopy, specifically the importance of the 2000–2500 nm band. We also see influences of the surveyed soil properties on the regression intercept (Figure 6b). The coefficient of pH is moderately large, negative, and precisely estimated. The coefficients of the soil texture attributes are more uncertain, which is not surprising since the attributes are highly correlated (Figure 2). The soil layer coefficients (Figure 6c) align with the fact that deeper soil layers have lower carbon concentrations (Figure 3). We also investigated the varying slopes of the LMM and found the variations were small (Figure S4) but contributed meaningfully to prediction performance (Figure S5).

4 | Discussion

4.1 | Main Findings

In this study we evaluated the transfer and uncertainty quantification performance of a variety of spectroscopy models in the context of a regional cluster of 8 agricultural farms. We compared existing models to a new approach: a linear mixed model (LMM) that leverages auxiliary information (farm, field, soil layer, and surveyed soil properties). In summary, we found that the LMM outperformed five existing approaches in both prediction performance and uncertainty quantification.

The prediction performance of the existing models varied; stratified PLSR and QRF did not improve on the PLSR baseline, while

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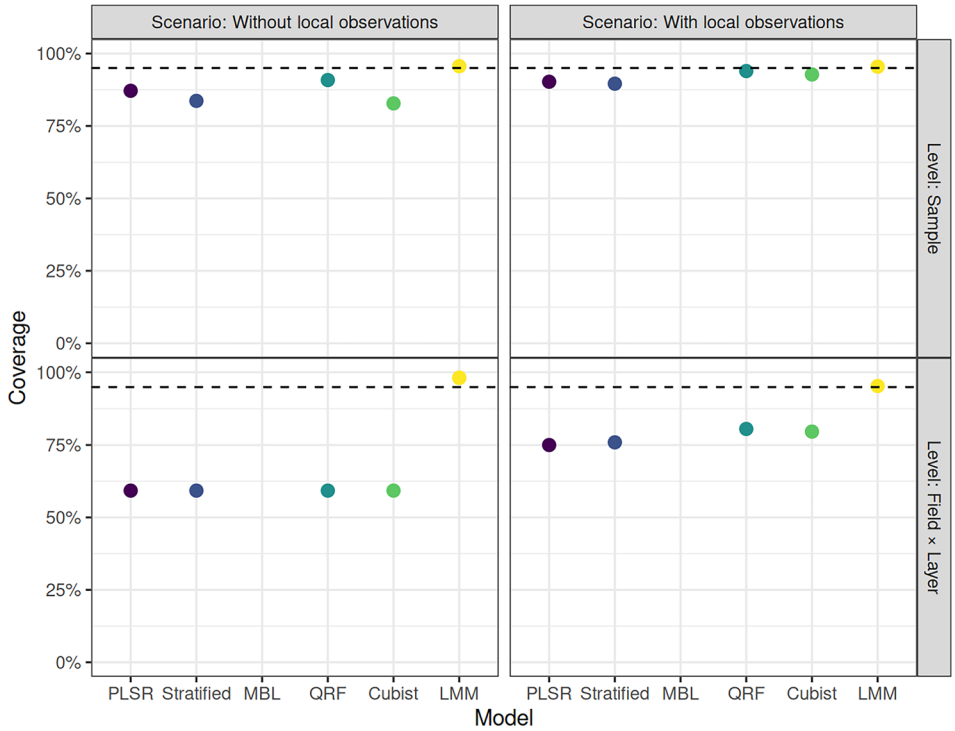


FIGURE 5 | Uncertainty quantification at (top row) sample and (bottom row) field-layer scale, that is one prediction for each soil layer (0–15, 15–30, 30–60 cm) in each field. Points show the empirical coverage rate of 95% prediction intervals compared to the nominal coverage rate of 95% (dashed line).

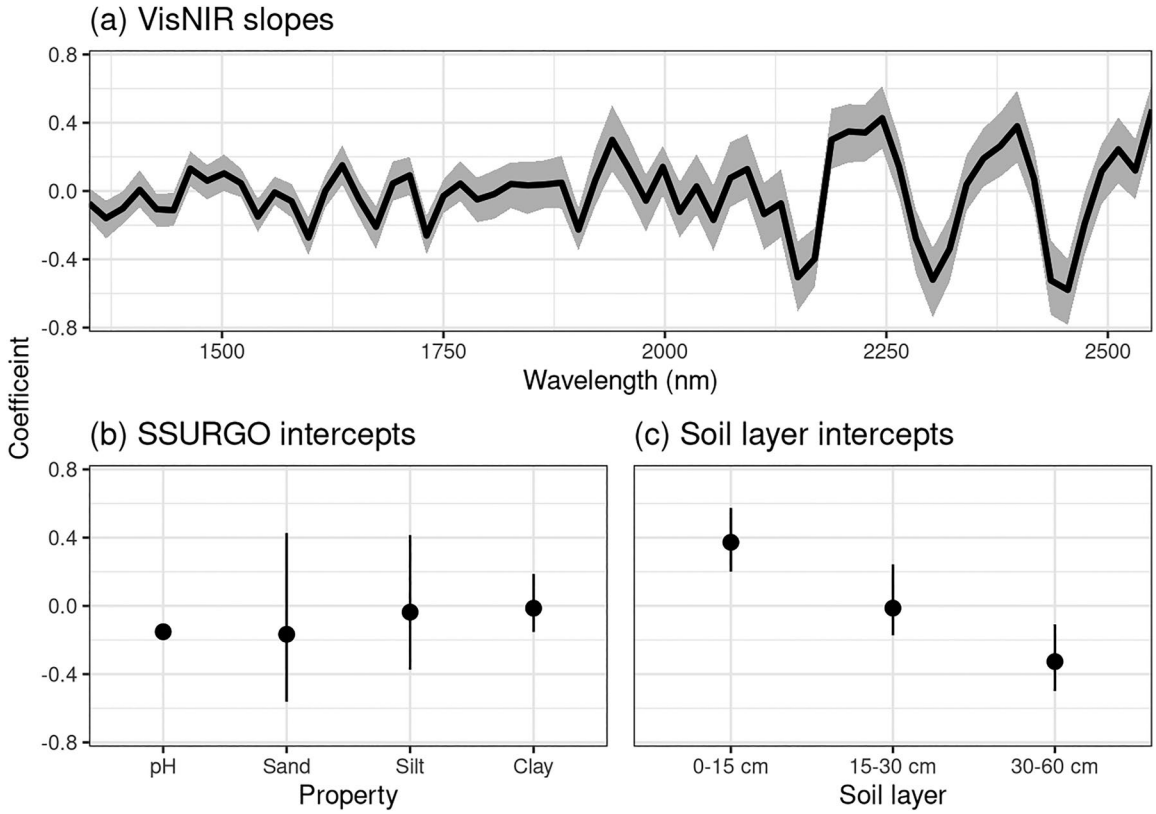


FIGURE 6 | LMM coefficients for (a) NIR reflectances, (b) field-level soil survey (SSURGO) predictors, and (c) soil layer intercepts. Note that outcome (on log1p scale), reflectances, soil properties have been standardized. All panels show Bayesian point estimates (posterior median) and uncertainties (posterior 50% interval). LMM, Linear mixed model; NIR, near-infrared; SSURGO, Soil Survey Geographic Database.

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MBL and Cubist performed better. However, the LMM consistently performed best, reducing RMSE compared to PLSR by an average of 27% and 10% without local observations and with local observations, respectively.

All models except MBL quantified uncertainty well for sample-level predictions. Specifically, they generated 95% intervals for these sample-level predictions with close to 95% coverage. However, when quantifying uncertainty for field-layer mean carbon concentration, existing models did not account for bias and severely underestimated the uncertainty of their field-layer predictions. The LMM was the only model accounting for correlated errors and so was the only model to correctly quantify uncertainty at the field-layer level (Figure 5).

4.2 | Considerations and Future Work

4.2.1 | Setting

The relatively small geographic scale (30 km) and homogeneous soils in our study make this a relatively favorable setting for a global PLSR model and therefore a relatively challenging setting for model transfer to improve on the PLSR baseline. Despite this, the LMM was a substantial improvement, suggesting the potential for even greater LMM transfer benefits in more heterogeneous soils (e.g., larger geographic scale).

4.2.2 | Dataset Size

In general, LMMs benefit from a large number of groups and samples per group in their training data (Gelman and Hill 2007). We had 8 farms, 18 fields, and 54 field-layers, with at least 3 samples per group. In the extreme cases of just one or two groups, or just one observation in each group, the LMM will be unable to estimate a group-level structure. However, the LMM should "fail gracefully" in these situations and reduce to a global regression. Future work should explore the sensitivity of LMM performance to the dimensions of training data. Sampling a greater number of farms than the eight included in this study will improve our understanding of between-farm variability in model parameters and performance.

4.2.3 | Group-Level Predictors

For the local regression for a particular group (e.g., farm), the LMM relies on: (i) observations in that group, if any are available in the training data; and (ii) group-level predictors in conjunction with observations from other groups. Without local observations, it is not surprising that the group-level predictors were crucial for LMM performance. Conversely, with local observations, that is including samples from the target farm in training, group-level predictors did not benefit LMM performance (Figure S6).

Without local observations, the availability and quality of group-level predictors will have a large effect on LMM performance. We used SSURGO estimates of texture and pH, which are widely available in the USA, though their quality and relevance for spectroscopy prediction may vary. More precise estimates of these variables could improve LMM performance. Future work could validate these predictors in other regions of the USA and for outcomes besides carbon concentration, as well as in other countries (from other sources) and consider other predictors altogether. However, we emphasize that in our study these variables were only important without local observations.

4.2.4 | Modeling Extensions

One benefit of our LMM is that it can be extended using well-developed Bayesian regression techniques. For example, a multi-outcome model (also known as multi-output or multi-task), in which multiple soil properties are modeled jointly. Specifically, a joint model of bulk density and carbon concentration would produce a model of carbon stocks, while accounting for potential residual correlation. Another example is a Gaussian process (kriging) component to account for residual spatial autocorrelation (e.g., Potash, Nakayama, et al. 2025). The random effects structure of the LMM could be expanded to include: core locations (useful for core-level aggregation or sampling designs where not all soil layers are measured for ground truth), finer geography (map units instead of fields), different years, different projects, and different sampling and measurement techniques. It is straightforward to implement these extensions (e.g., using the brms R package). However, the LMM in this study already requires considerable computation (~15 min on a budget laptop), though a simpler LMM (random intercepts but not slopes) performed almost as well (Figure S5) in seconds.

ML models can be extended as well, for example, to include the LMM features of (i) a grouped structure of farms, fields, and soil layers and (ii) auxiliary group-level predictors. Including these variables in Cubist improved performance (Moura-Bueno et al. 2021), though the LMM still performed best (Figure S7). To handle correlated errors, conformal prediction could be extended to include a group structure, though to our knowledge this is not standard. More generally, while many extensions to ML models are possible, they may be less straightforward than for Bayesian regression.

4.2.5 | Modeling Philosophy

While our LMM coefficients vary across farms, fields, and soil layers, this variation is parameterized by linear relationships with group-level predictors and normally distributed residuals. Compared to less parametric ML, this parameterization may seem overly constrained. However, the LMM performed best in our study and is successful in many other applications. Moreover, the LMM is often easier to interpret (Figure 6) than black-box ML.

Our LMM is fit on the entire training set with Bayesian (MCMC) inference. This is unlike ML models which rely on internal cross validation to optimize hyperparameters or obtain residuals for conformal prediction. Cross validation is problematic if the training data is so small that each observation contributes significantly to model fit. It is also a computational burden for models with a large number of hyperparameters.

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ML often produces pipelines composed of modular components (e.g., conformal prediction), whereas Bayesian modeling favors unified probability models. As a result, in our experience, ML pipelines can be more computationally efficient and easier to develop (if the desired components exist, see previous section); whereas Bayesian models excel at uncertainty quantification and data efficiency. Of course, a Bayesian model such as our LMM could itself be a component in an ML pipeline– a direction for future research.

5 | Conclusion

Two challenges that limit the adoption of soil spectroscopy are model transfer and uncertainty quantification. In this study, we collected a novel dataset of traditional soil carbon measurements as well as NIR spectra in a cluster of 18 agricultural fields in Northeastern USA. We evaluated transfer and uncertainty for a variety of existing approaches as well as a new approach that we proposed using a linear mixed model (LMM) that leverages pre-existing soil survey data.

We found that our LMM outperformed existing approaches in (i) model transfer and (ii) uncertainty quantification of aggregated predictions. This strong performance, combined with a proven track record in many other applications, makes LMMs a promising approach for soil spectroscopy.

Author Contributions

Eric Potash: conceptualization, methodology, software, writing – original draft, writing – review and editing. **Colleen Partida:** data curation, writing – review and editing. **Jose Safanelli:** methodology, writing – review and editing. **Elizabeth Forbes:** investigation, writing – review and editing. **Alex Polussa:** data curation, writing – review and editing. **Tiffany Runge:** investigation, writing – review and editing. **Matthew Sheffer:** investigation, writing – review and editing. **Jonathan Sanderman:** writing – review and editing, supervision.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data and code that support the findings of this study are available in figshare at https://doi.org/10.6084/m9.figshare.34059552.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Figure S1:** Comparison of prior and posterior distributions for key parameters. **Figure S2:** Ratio of posterior to prior distribution variance for all 5731 model parameters. Smaller values indicate that the prior is less informative. **Figure S3:** Winkler scores (lower is better) at (top row) sample and (bottom row) field-layer scale. Points and lines are mean and standard error, respectively. **Figure S4:** (a) Variation in slopes by field-level SSURGO predictors and (b) standard deviation of random effects (intercepts and slopes) by grouping level. **Figure S5:** Influence of varying slopes on LMM performance. Fixed slope (varying intercept) LMM does not perform as well as varying slope LMM, but performs better than Cubist. Note that the fixed slope model can be fit about 100 times faster (9 s vs. 900 s). **Figure S6:** Influence of SSURGO group-level predictors on LMM performance. SSURGO is important for performance in no local observations scenario but not important in local observations scenario. **Figure S7:** Influence of auxiliary information on Cubist performance. Adding field ID, soil layer, and SSURGO texture and pH predictors to Cubist improves performance in the local observations scenario but not in the no local observations scenario. In both cases, LMM performs best.

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